Check for updates

ACM DIGITAL LIBRARY | Association for Computing Machinery | acm open

Latest updates: https://dl.acm.org/doi/10.1145/3645104

PDF Download
3645104.pdf
28 February 2026
Total Citations: 61
Total Downloads: 2594

Published: 09 April 2024
Online AM: 08 February 2024
Accepted: 31 January 2024
Revised: 17 December 2023
Received: 07 September 2022

Citation in BibTeX format

SURVEY

Blockchain Data Storage Optimisations: A Comprehensive Survey

JUN WOOK HEO, Queensland University of Technology, Brisbane, QLD, Australia

GOWRI SANKAR RAMACHANDRAN, Queensland University of Technology, Brisbane, QLD, Australia

ALI DORRI, Queensland University of Technology, Brisbane, QLD, Australia

RAJA JURDAK, Queensland University of Technology, Brisbane, QLD, Australia

Open Access Support provided by:

Queensland University of Technology

Blockchain Data Storage Optimisations: A Comprehensive Survey

JUN WOOK HEO, Queensland University of Technology, Brisbane, Australia
GOWRI SANKAR RAMACHANDRAN, Queensland University of Technology, Brisbane, Australia
ALI DORRI, Queensland University of Technology, Brisbane, Australia
RAJA JURDAK, Queensland University of Technology, Brisbane, Australia

Blockchain offers immutability, transparency, and security in a decentralised way for many applications, including finance, supply chain, and the Internet of Things (IoT). Due to its popularity and widespread adoption, it has started to process an enormous number of transactions, placing an ever-growing demand for storage. As the technology gains more popularity, the storage requirements of blockchain will increase, necessitating storage optimisation solutions. Proposed solutions for blockchain storage efficiency range from reducing the degree of data replication to redacting or compressing data. Each of these storage optimisation categories involves a complex interplay with the timing of blockchain data processing and mining, yet no existing survey analyses these dimensions. This article surveys the state-of-the-art blockchain storage optimisations and categorises them into replication-based, redaction-based, and content-based optimisations. Replication-based optimisations focus on reducing duplication of blockchain data shared among participants after committing data on the blockchain ledger. Redaction-based optimisations allow users to modify or delete data already committed on the ledger in various ways, while content-based optimisations compress data before or after committing it to the ledger. We analyse and evaluate these solutions in the aspects of security, decentralisation, and scalability. We present the advantages and disadvantages of the existing blockchain storage optimisations and comprehensively compare them. Additionally, we discuss the opportunities and challenges for future work to optimise blockchain storage.

CCS Concepts: • **General and reference → Surveys and overviews**; • **Networks → Peer-to-peer protocols**; • **Information systems → Distributed storage;**

Additional Key Words and Phrases: Blockchain, blockchain storage optimisation, blockchain scalability

ACM Reference Format:
Jun Wook Heo, Gowri Sankar Ramachandran, Ali Dorri, and Raja Jurdak. 2024. Blockchain Data Storage Optimisations: A Comprehensive Survey. *ACM Comput. Surv.* 56, 7, Article 179 (April 2024), 27 pages. https://doi.org/10.1145/3645104

1 INTRODUCTION

Blockchain is a distributed database that is shared and managed by all participants in a peer-to-peer network. Numerous application domains, including finance, **Internet of things (IoT)**, or supply chains use the blockchain technology because of its inherent support for immutable storage,

Authors' address: J. W. Heo, G. S. Ramachandran, A. Dorri, and R. Jurdak, Queensland University of Technology, 2 George Street, Brisbane, Queensland, Australia, 4000; e-mails: junwook.heo@hdr.qut.edu.au, g.ramachandran@qut.edu.au, ali.dorri@qut.edu.au, r.jurdak@qut.edu.au.
Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
© 2024 Copyright held by the owner/author(s).
ACM 0360-0300/2024/04-ART179
https://doi.org/10.1145/3645104

transparency, cryptocurrency, decentralised computation through smart contracts, and security [18, 23, 57, 70, 96] . Many real-world blockchain applications produce enormous amounts of data, resulting in a significant increase in transactions and computational costs. This limitation poses challenges to scalability and sustainability of blockchain. Vitalik Buterin, the founder of Ethereum, points out the tradeoffs between three critical properties: decentralisation, security, and scalability, the so-called scalability trilemma. Buterin and others in the blockchain community argue that a blockchain cannot guarantee these three properties at the same time [1, 9]. Researchers have proposed practical scalability solutions balancing the tradeoffs by changing the blockchain architecture, improving the consensus algorithm, or reducing the storage space [17, 36, 66, 102].

The rapid growth of the ledger and the tremendous demand for data purging against blockchain immutability have led to active research on blockchain storage optimisation. In 2023, the ledger sizes of Bitcoin and Ethereum reached 500 and 700 GBytes, respectively, and continue to grow over time [10]. In addition, there is a demand for blockchain data modification scrutinising the immutability of blockchains. The **European General Data Protection Regulation (GDPR)** regulates how personal data is handled by declaring the "right of rectification" and "right to be forgotten," so strict requirements are placed on service providers for the processing of the personal data [37, 60, 76, 80]. Another issue with the immutability of blockchain is privacy-sensitive or harmful data intentionally or unintentionally stored on the blockchain [27, 42, 52, 75, 95, 99]. For example, according to Reference [53], Bitcoin contains more than eight files with illegal and condemned content, and seven files violate copyrights. The users may want to delete their personal data and the system may also need to remove the harmful data from the blockchain not only to save the storage space but also to improve security and privacy.

Researchers and blockchain platform developers proposed numerous approaches to manage and optimise blockchain storage to cope with the growing ledger size and the demands for data redaction. Existing surveys that review the state-of-the-art on blockchain storage fail to classify the existing solutions into different categories and focus mainly on ways to minimise storage [8, 77, 90]. Benisi et al. [8] survey the decentralised storage networks and concentrate primarily on four storage solutions, i.e., Storj, Filecoin, Sia, and Swarm. Besides, Reference [8] fails to dissect the existing solutions while studying how consensus and cloud storage can address the blockchain storage problem. Xie et al. [90] survey the overall scalability of blockchain systems in a high-level magazine article by focusing on all aspects of scalability, including the consensus algorithm, storage, and networking. Their work does not provide a detailed survey targeted at storage optimisation. Sharma et al. [77] survey how the blockchain technology can be used for cloud storage, further exacerbating the scalability and the storage problem. To the best of our knowledge, existing surveys lack the following:

— Comprehensive survey of blockchain storage optimisation solutions
— Categorisation of blockchain storage optimisation solutions
— Dissection of existing storage solutions to explain what operations get executed at the preprocessing, mining, storage, and post-processing phases
— In-depth analysis of the factors underpinning blockchain scalability within the blockchain trilemma, namely, network efficiency, computational efficiency, and storage efficiency.

This article surveys existing blockchain storage optimisations and classifies them into three categories: (i) Replication-based, (ii) Redaction-based, and (iii) Content-based optimisation. Figure 1 shows blockchain layers where these blockchain storage optimisation solutions operate. The components in the storage layer play a crucial role in managing the blockchain data that is committed after mining. These components handle the storage and retrieval of data within the blockchain system. However, the components in the **Blockchain (BC)** layer form the foundation

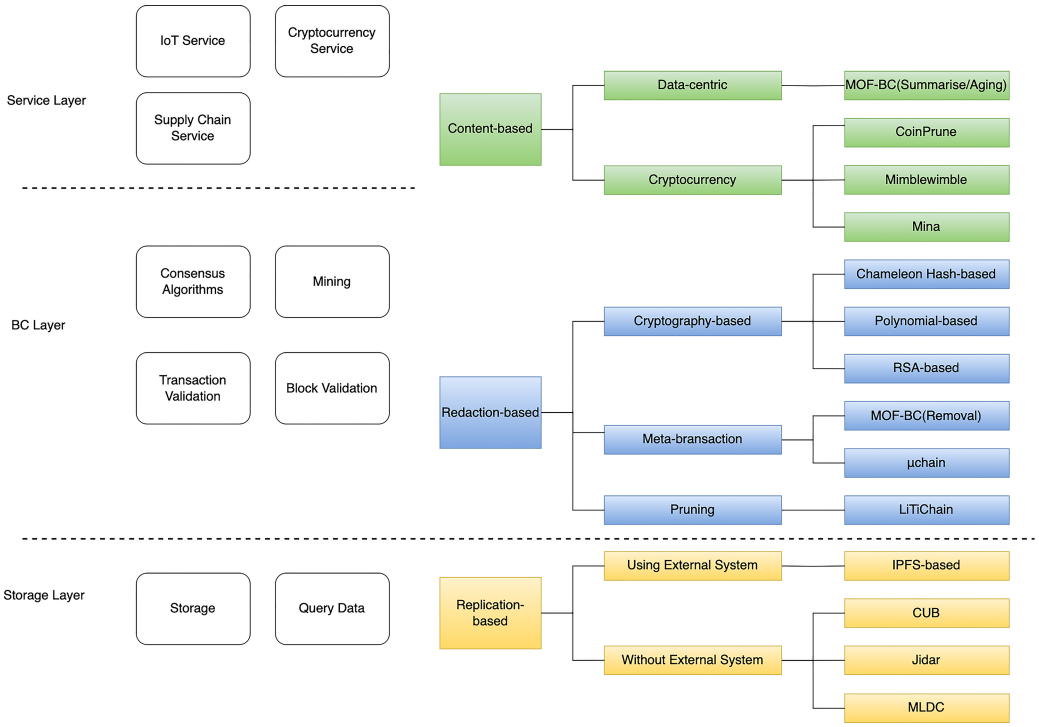


Fig. 1. Blockchain storage optimisations.

of the blockchain protocol itself. They encompass the core functionalities and mechanisms that enable consensus, validation, and transaction processing within the blockchain network. Last, the service layer components represent a diverse range of services that utilise blockchain networks. These components provide higher-level functionalities and interfaces for applications, such as IoT service, cryptocurrency service, supply chain service, and other services built on top of the blockchain infrastructure [94, 96, 97]. We provide a more detailed description of blockchain storage optimisation categories below:

— Replication-based optimisations operate at the storage layer, meaning the solutions aim to enhance storage efficiency by reducing the number of replications already committed on the blockchain ledger among storage nodes. Note that replication in the blockchain storage context refers to keeping multiple copies of data in different nodes in a distributed network, whereas the replication-based optimisation techniques focus on *minimising the number of data duplicates without any changes to the ledger*. As a result, reducing the number of copies tends to compromise data availability, so replication-based optimisations introduce their own solution to this problem to maintain the entire ledger.
— One of the essential characteristics of blockchain is immutability. However, there are significant demands for a mutable blockchain that addresses security, privacy, and storage efficiency concerns. In this article, we focus on the ability of redactable blockchains to reduce the data size of blockchain as claimed by many solutions [15, 25, 67]. Redaction-based optimisations seek to reduce the size of the ledger itself in various ways, such as modifying, deleting, and pruning data across blockchain layers. This approach mainly focuses on modifying data that is already committed to the blockchain ledger by *updating the ledger itself*.

Redaction-based approaches are generally agnostic to the content type of data stored in blockchains.
— Content-based optimisations, similar to redaction-based optimisations, aim to reduce the size of the ledger, yet they differ from redaction-based optimisations by focusing on the content type. This approach utilises various compression techniques, including lossless or lossy compression, to achieve efficient storage. Additionally, novel approaches have been proposed to address the challenge of reducing transaction history for cryptocurrency blockchains.

We clarify the taxonomy of the blockchain storage optimisations based on how blockchain data is optimally stored and where storage optimisation solutions operate in the blockchain architecture. We also perform an extensive qualitative analysis based on the aforementioned scalability trilemma to present a detailed comparison of existing blockchain storage optimisation solutions.

This study makes the following contributions:

— A new taxonomy of blockchain storage optimisation solutions. We classify existing blockchain storage optimisations into three categories according to the implemented layers and optimisation methods.
— An extensive analysis of the state-of-the-art literature with regard to blockchain storage optimisation solutions. We examine the tradeoffs between decentralisation, security, and scalability of existing solutions through a comprehensive literature review.
— A detailed comparison of existing blockchain storage optimisation solutions. We analyse and evaluate existing solutions with performance matrices considering the blockchain scalability trilemma. We also consider the factors underpinning blockchain scalability within the blockchain trilemma, namely, the tradeoffs among network efficiency, computational efficiency, and storage efficiency.

This article primarily focuses on surveying blockchain optimisation solutions that aim to reduce the data size of the blockchain ledger and minimise the number of replications. The majority of these approaches specifically address the storage challenges associated with the traditional blockchain architecture, where blocks are chained to previous blocks. To the best of the authors' knowledge, no storage optimisation solution has targeted **Directed Acyclic Graph (DAG)**-based blockchains like IOTA [64, 78], so we exclude them from the survey in this article. Additionally, metadata optimisations for stateful blockchains like Ethereum are also excluded from the discussion. These excluded topics can be considered as potential areas for future research.

2 OVERVIEW OF BLOCKCHAIN

This section outlines a background of the blockchain system including the fundamental architecture and key features based on Bitcoin blockchain. Sections 2.1, 2.2, and 2.3 outline the details of transaction, block, and mining, respectively. Subsequently, Section 2.4 discusses the blockchain storage nodes of Bitcoin.

Blockchain is a distributed, immutable, and secure ledger with a sequence of blocks managed by participants in peer-to-peer networks without any centralised authorities. Miners generate blocks including a list of transactions through various consensus mechanisms such as **Proof-of-Work (PoW)** or **proof-of-stake (PoS)**. Each block includes a hash of the previous block in its header to ensure that the blockchain is not tampered with. Consensus mechanisms such as the longest chain rule also provide a way to reach agreement on a single state of a decentralised network, so they keep the blockchain secure [55, 56].

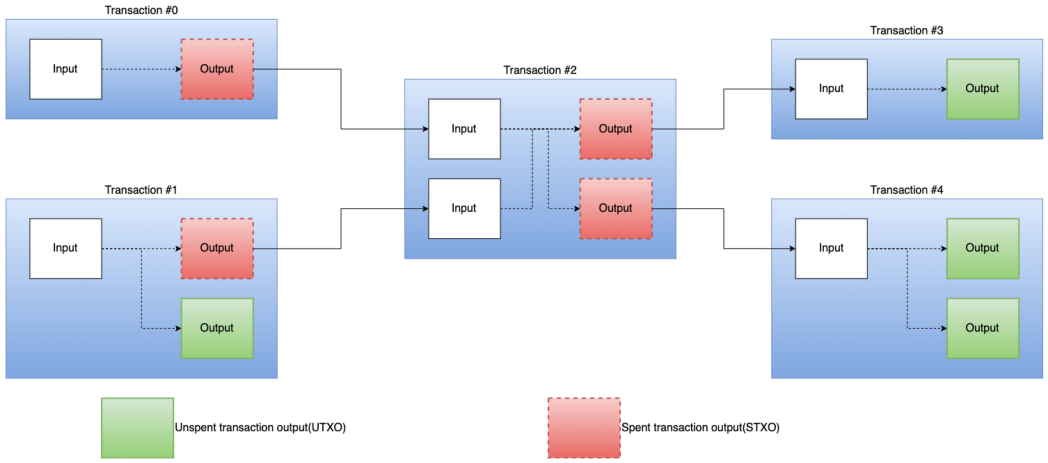


Fig. 2. An example of transaction chain.

2.1 Transaction

A transaction is a basic data structure consisting of inputs and outputs that are recorded on the blockchain ledger. Figure 2 shows an example of the Bitcoin transaction chain. A transaction output is used as a transaction input to represent the amount of currency and the ownership of the currency. If an output is used as an input to another transaction, then it is marked as **Spent Transaction Output (STXO)** and can no longer be input to another transaction. The outputs in Transaction #0 and #1 are, for example, spent by Transaction #2 and the outputs in Transaction #2 are referenced by Transaction #3 and #4. Otherwise, the output remains as **Unspent Transaction Output (UTXO)**, which the owner can spend in the future transaction. Every transaction changes the UTXO set in the Bitcoin system and full nodes can trace those UTXOs [14, 21, 47].

In Bitcoin, a transaction output consists of an amount of Bitcoin and a locking script, so-called scriptPubKey. However, a transaction input consists of a transaction ID and unlocking script, which is known as scriptSig. The transaction ID indicates the transaction in which the UTXO is spent. The scriptPubKey in a transaction output specifies the condition to spend the output in the future by including the hash of the public key of the recipient who will own the Bitcoin in the output, whereas the scriptSig in a transaction input is a script to unlock the condition locked by the scriptPubKey [22, 30, 61, 83].

Unlike Bitcoin, transactions in Ethereum represent valid information that can transition between states. As a stateful blockchain, the world state of Ethereum is stored in the state root in the block header, maintained by a modified Merkle Patricia tree. Please refer to the Ethereum yellow paper [89] for details.

2.2 Block

A block is identified by a hash using cryptographic hash algorithm and consists of a block header and transactions in Bitcoin. As shown in Figure 3, the block header contains a previous block hash, timestamp, Merkle root, difficulty target, and nonce. The previous block hash makes the blockchain data structure an ordered and back-linked immutable list of blocks. Difficulty and nonce are used by miners to compute a block hash when creating new blocks [32, 62, 83].

The Merkle root represents an overall digital fingerprint of the transaction list in a block to verify the inclusion of a transaction within a block. The Merkle root is calculated by recursively hashing

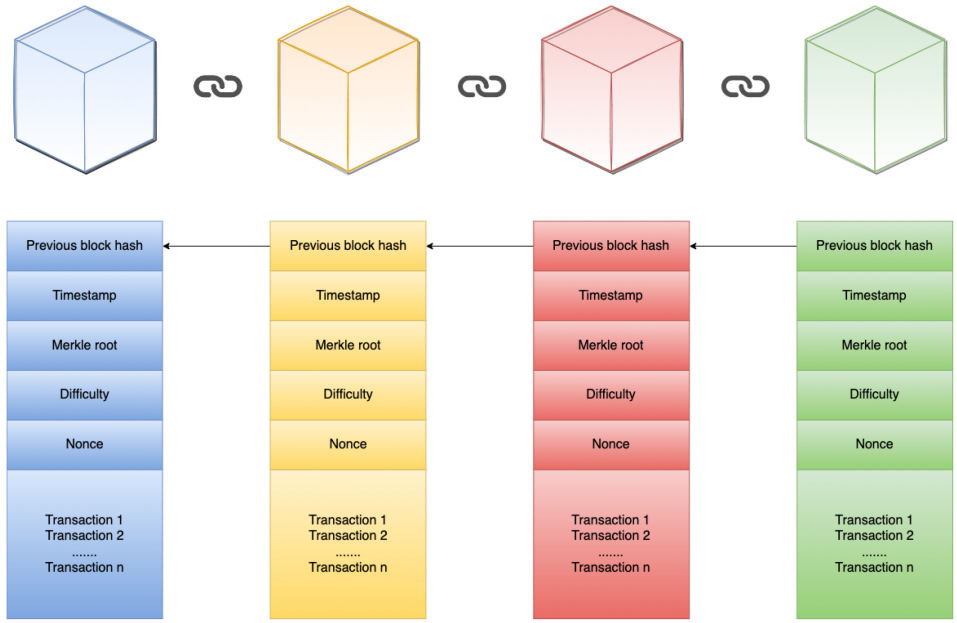


Fig. 3. Blockchain structure.

pairs of transactions until there is only one hash left in the Merkle tree. Figure 4 illustrates how to use a Merkle tree to verify the inclusion of a data element. For example, anyone can easily prove that the Merkle root $H_{abcdefgh}$ contains H_c in the orange box by computing three hashes. First, compute H_{cd} and H_{abcd} in the dashed boxes using the hashes H_d and H_{ab} in the blue boxes. Then, compute $H_{abcdefgh}$ in the dashed box using the hash of H_{efgh} in the blue box [31]. This mechanism is utilised for **Simplified Payment Verification (SPV)** without downloading all transactions in a block [3, 46, 84].

2.3 Mining

Mining underpins the decentralisation and security of the blockchain by verifying the legitimacy of transactions without centralised authority. In Bitcoin, for instance, miners validate transactions and add them to a candidate block. Then, a new block is generated every 10 minutes on average by a miner who solves a cryptographic puzzle called PoW. Finally, the new block is shared among all participants in the network.

It is difficult to achieve decentralised consensus without a central authority in a peer-to-peer network. To address this issue, all full nodes in Bitcoin perform independent verification for each transaction and aggregate the verified transactions into a candidate block. New blocks generated by computation of PoW algorithms are also independently verified and assembled into a chain by every node. A block is considered final following the longest chain rule. In practice, a block is considered final when six blocks are appended to it [45, 86, 103].

PoW of Bitcoin produces a hash of a new block by changing the nonce according to the difficulty target. A miner calculates the hash of a candidate block containing verified transactions and the block header. If the computed hash is greater than the difficulty target, then the miner usually increases the nonce by one and calculates the hash again. The difficulty of the Bitcoin network is adapted to keep the block generation period at 10 minutes [28, 98, 100].

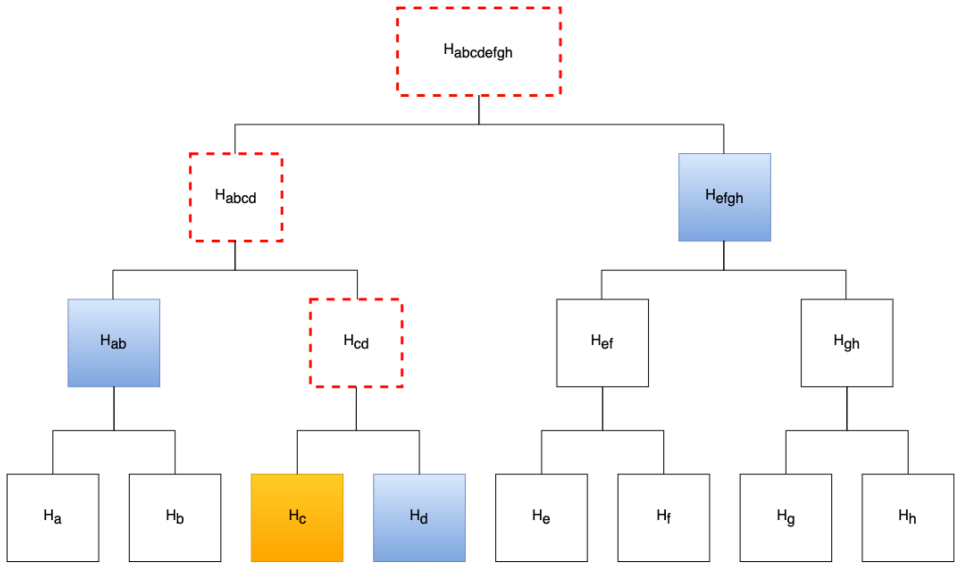


Fig. 4. An example of Merkle tree: The orange box indicates the hash of a transaction that needs to be proven to be in the Merkle tree. The blue boxes indicate known hashes of transactions in the tree, while the boxes with the dashed red outline are hashes that can be computed based on the known hashes.

2.4 Storage in Immutable Ledger

After the mining process, the successful miner proposes the next block to be added to the blockchain. The nodes in the distributed network must receive the block broadcasted by the successful miner and append it to the blockchain after verifying the block's integrity. Here, the nodes are classified into two categories: full and light nodes. The Bitcoin protocol recommends that the full nodes store the entire blockchain, including the block headers and the transactions in each block, whereas the light nodes store only the header of the blocks. The security and decentralisation of blockchain networks heavily rely on maintaining a robust population of full nodes. Therefore, any reduction in the number of full nodes can potentially compromise the security and decentralisation aspects [27].

3 STORAGE STATISTICS OF CONTEMPORARY AND REPRESENTATIVE BLOCKCHAINS

Dozens of blockchain platforms are actively serving millions of users globally. This section reviews the storage statistics of the three most popular and representative blockchain platforms to highlight the growing storage demands. We picked Bitcoin, Ethereum, and Litecoin due to their popularity and age. Figure 5 shows the storage cost of Bitcoin, Ethereum, and Litecoin. The graph illustrates the following statistics:

— Total block size (Byte): Sum of total block size (accumulated)
— Avg block size (Byte): Average block size per year
— Avg number of transactions per block: Average number of transactions per block for a year.

Note that the statistical data presented in this section is obtained through the APIs provided by Reference [10]. We will discuss each of these blockchain platforms in the rest of this section.

3.1 Storage Status of Bitcoin

Bitcoin started the blockchain revolution on January 3, 2009, by mining the first block. Since its inception, the number of blocks has been increasing [4, 43, 59]. The total number of blocks in Bitcoin as of July 2023 is 799,579. Tens of thousands of blocks include millions of transactions over a decade. All these transactions and block data are critical for the operation of Bitcoin, as each transaction must consist of a sender and receiver, wherein the sender must have sufficient Bitcoins in their account to make a transfer to a receiver's account. Bitcoin's protocol includes a **simplified payment verification (SPV)** that relies on the information stored on the blockchain. Hence, the blocks must be stored on the nodes that drive the Bitcoin network. However, the cost of storing the Bitcoin blockchain increased dramatically over the decade, as shown in Figure 5. The block size has doubled in the past four years, rising from 250 GB to 500 GB. Such growth demands high storage capacity in the network, meaning each full node must be able to retain several hundred GB of Bitcoin blockchain. Due to the distributed nature of Bitcoin, multiple nodes must retain the entire blockchain, introducing high storage costs to the full nodes. Besides, average block size and the number of transactions in a block are also tuned over time to increase the throughput of Bitcoin, as shown in Figure 5. These results highlight the need to manage the storage cost of Bitcoin.

3.2 Storage Status of Ethereum

Ethereum is one of the most popular blockchain platforms after Bitcoin, because it supports smart contracts, allowing application developers to create multiple decentralised applications, including tokens. It also follows the PoW protocol to create blocks and also rewards the successful miner with *Eth* cryptocurrency. Note that there are developments in the Ethereum ecosystem on the **Proof-of-Stake (PoS)** protocol, an alternative to PoW, which aims to move away from the expensive mining process of PoW. Miners proposed the first block of Ethereum on July 30, 2015 [16, 81]. At present, there are 17,738,278 blocks in Ethereum. Figure 5 shows the evolution of the Ethereum blockchain. Although Ethereum started six years after Bitcoin, it has roughly 700 GB of transactions in the blockchain, demanding almost the same storage as Bitcoin. Smart contracts could be the reason behind this rapid growth of Ethereum, as several decentralised applications are developed using smart contracts on Ethereum. The block size and number of transactions in the block are adjusted to enhance the throughput and the scalability of Ethereum, as shown in Figure 5. These storage trends of Ethereum further motivate the need to manage its storage, as a regular user may not be able to afford significant storage hardware to be in the Ethereum network, threatening its decentralisation.

3.3 Storage Status of Litecoin

Litecoin was launched on October 7, 2011, with its first block. It follows a similar protocol as Bitcoin but with a modified Bitcoin code. The critical changes to Litecoin include using the script hashing algorithm in place of Bitcoin's SHA-256 and a shorter block interval of approximately 2.5 minutes compared to Bitcoin's 10 minutes block interval. These changes attracted community members to use this blockchain platform for financial applications [33, 35, 63]. The total number of blocks in Litecoin sits at 2,512,709 as of July 2023. It is important to note that over 2.5 million blocks are produced in a little over a decade, demanding approximately 120 GB of storage as of July 2023, as shown in Figure 5. The average block size remained at 36 KB until 2018 and rapidly increased to 280 KB in 2023, highlighting the rising demand for storage. Similarly, the number of transactions in the block quickly increased between 2018 and 2023. All these changes highlight the changes made to the underlying protocol to meet the throughput and scalability demands, but it, unfortunately, comes at the cost of high storage.

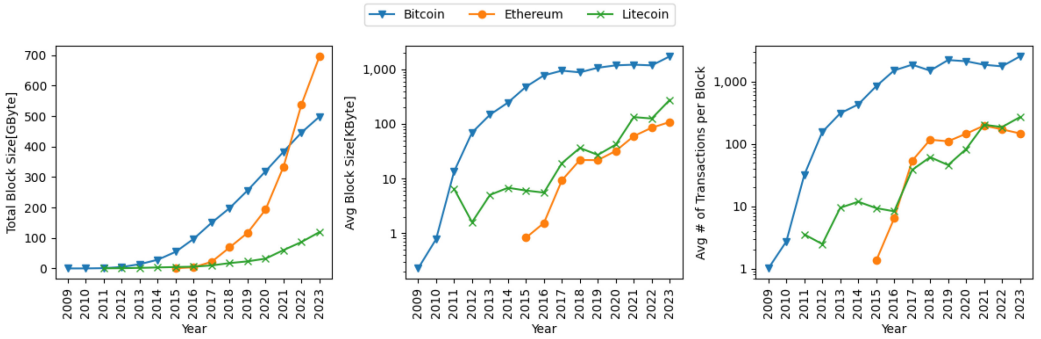


Fig. 5. Blockchain storage statistics.

3.4 Summary

The review of the evolution of Bitcoin, Ethereum, and Litecoin blockchains and their storage demands emphasises the need to manage the storage. It is important to note that the public nodes are essential to upholding the decentralisation property of the blockchain. For the public nodes to participate in the blockchain's mining and storage processes, the platforms need to have manageable storage demands, as regular users cannot afford expensive computing and storage platforms. Figure 5 also shows that the protocols are undergoing multiple reconfigurations to handle more transactions. Still, such changes inadequately consider the storage demands, questioning the sustainability of blockchain technology. In the rest of the article, we review the storage management and optimisation solutions in the literature and industry to help the researchers and industry stakeholders better manage the alarming storage challenge.

4 DISTRIBUTED STORAGE

Contemporary blockchains such as Ethereum and Bitcoin require the full nodes that perform the mining process also to store the entire ledger. However, the size of the blockchain ledger has grown so large that it poses challenges for standard full nodes to store the entire ledger, resulting in inadequate full nodes. As a solution to this challenge, some blockchain optimisations utilise distributed storage schemes, such as the **InterPlanetary File System (IPFS)**, for storing blockchain data. Various distributed storage schemes exist in both literature and industry, offering optimisation solutions for blockchain storage as external storage in a decentralised and secure manner. The remaining part of this section will provide a detailed discussion of these multiple distributed storage schemes:

— InterPlanetary File System (IPFS) [6] is a decentralised peer-to-peer distributed file system using content addressing and Merkle **Directed Acyclic Graphs (DAGs)** to identify and manage contents. However, it does not guarantee file persistence. The data availability is not highly reliable and the storage efficiency is also low, because replication strategies are less efficient than using erasure codes in terms of data availability and storage efficiency.
— Storj [88] is a decentralised storage framework that utilises encryption, shards, and data distribution across nodes around the world. Storj enhances the level of availability by using erasure codes. In terms of security, the audit and reputation mechanisms enhance the protection of data. However, erasure coding increases computational costs. In addition, when dealing with a small size of data, storage overhead can increase due to the padding data.
— Swarm [82] is a permissionless and decentralised storage based on an existing smart contract platform, Ethereum, to implement a financial incentive system. Swarm proposes a

Distributed Immutable Store for Chunks (DISC), which contains the information about the content itself instead of using DHT. Compared to Storj, the storage efficiency of Swarm is slightly lower, because it mainly uses replication to improve availability instead of using erasure codes with sufficient redundancy.

— Sia [85] is a decentralised storage platform that allows users to enter into storage contracts with their peers. Contracts between storage providers and clients define the data to be stored and the price for storage. To maintain the contracts, Sia also proposes its own proof of storage scheme. Sia improves security through the proof of storage and encryption of data. Using erasure codes improves availability while it also increases storage overhead for the small size of data due to high storage overhead.

— Filecoin [7] is a distributed storage network working as an incentive component on top of IPFS. To ensure reliable storage, Filecoin suggests Proof-of-Replication and Proof-of-Spacetime using **Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (zk-SNARK)**, which enhances security and data availability. Filecoin also uses data encryption and erasure coding. Although data availability is improved over IPFS, Filecoin does not guarantee data persistence.

Why optimise blockchain storage? Existing distributed storage schemes have various limitations, including availability and high storage cost. Addressing these limitations is challenging, as the prior research around the blockchain trilemma shows that it is impossible to guarantee scalability, decentralisation, and security at the same time [1, 9]. It also applies to blockchain storage.

5 EVALUATION PARAMETERS FOR STORAGE OPTIMISATIONS

Blockchain storage optimisation solutions primarily focus on scalability as the primary design target. We analyse the impacts on security, decentralisation, and scalability.

5.1 Metrics for Evaluation

5.1.1 Security. In terms of security, the **Confidentiality, Integrity, and Availability (CIA)** triad captures the fundamental elements of information security [74]. Based on the CIA triad, we evaluate the impact of blockchain storage optimisations on security using the following metrics:

— **Confidentiality (C):** This metric assesses the protection of information from unauthorised access.

— **Integrity (I):** This metric evaluates the protection of information from unauthorised modification.

— **Availability (A):** This metric measures the accessibility of information to authorised users.

5.1.2 Decentralisation. Many optimisation solutions compromise decentralisation by adding central authorities or concentrating data or traffic on a few nodes. We evaluate the impact of these optimisation solutions on the decentralisation using the following metrics:

— **Decentralised Management (DM):** This metric evaluates the level of decentralisation in optimisation, indicating that there is no central authority responsible for managing the optimisation scheme.

— **Degree of Decentralisation (DD):** This metric assesses the degree of decentralisation in optimisation by evaluating how well it maintains decentralisation in terms of resource concentrations, such as network, computation, or storage.

5.1.3 Scalability. The primary aim of blockchain storage optimisation solutions is to enhance storage efficiency, but they can also affect network efficiency and computational efficiency. We

evaluate the impact of these optimisation solutions on these scalability factors using the following metrics:

- **Network Efficiency (NE):** This metric measures the impact of the optimisation on network overhead, which includes increased network traffic and query costs. Query cost is defined as the additional cost for the remote requests, as the cost for retrieving data from a local storage is insignificant. Network efficiency evaluates any impact on the blockchain network such as transaction throughput or latency due to the optimisation.
- **Computational Efficiency (CE):** This metric quantifies the additional computing power required due to the optimisation. It takes into account the increased remote queries, data verification processes, additional processes for data removal, and any additional computations needed for redacting or compressing data in the blockchain ledge.
- **Storage Efficiency (SE):** This metric evaluates the amount of storage space reduced due to the optimisations.

By considering these evaluation parameters, we will thoroughly assess the impact of replication, redaction, and content-based optimisations on security, decentralisation, and scalability in blockchain systems in Sections 6, 7, and 8.

5.2 Summary of Existing Optimisations

We categorise blockchain storage optimisation into the three categories of replication-based, redaction-based, and content-based schemes that aim to enhance blockchain scalability. As shown in Table 1, we identify advantages and disadvantages of existing blockchain storage optimisations. Replication-based approaches tend to have favourable storage efficiency by reducing the number of duplicates. However, in achieving this, they often sacrifice either data availability or network efficiency. However, redaction-based optimisations provide the way to remove or update data committed to the ledger while they tend to sacrifice security or decentralisation. Most content-based optimisations focus on reducing cryptocurrency transaction records, which makes them more application- or platform- specific. The above discussion highlights that each category of blockchain storage optimisation typically trades off one goal of the blockchain trilemma for another. Further, it shows the tradeoffs within the scalability goal, such as network efficiency, computational efficiency, and storage efficiency, in certain approaches. For this reason, we expand the blockchain trilemma definition so the scalability goal is segmented into network, storage, and computational efficiency. In the next section, we review the existing solutions for each category and explore the details of their pros and cons in light of the expanded blockchain trilemma. Because the scalability metrics are affected by different factors for each category, we provide the category-specific definitions of each of the scalability metrics (network, storage, and computational efficiency).

6 REPLICATION-BASED OPTIMISATIONS

Replication-based solutions optimise replication of blockchain data committed to the ledger after mining, so they do not provide a way to remove or update data that has already been committed to the ledger. There are two types of approaches. One uses external systems to store data such as IPFS (discussed in Section 6.1), while the other stores data in the local storage of participant nodes (discussed in Section 6.2). Since replication-based optimisations mainly play a role in distributed storage, it is important to maintain a high level of data availability that is balanced with storage efficiency.

Replication vs. replication-based optimisation: Replication in the context of blockchain storage refers to duplicating blocks, meaning multiple copies are retained on numerous nodes in the network, maximising data availability. *Replication-based optimisation*, however, refers to

Table 1. Summary of Existing Blockchain Storage Optimisations

Optimisations		Advantage	Disadvantage
Replication-based	**IPFS-based** [44, 101]	1. Keep blockchain hash consistency 2. High storage efficiency	1. Increase traffic for transaction 2. Using external resources 3. Concentrate traffic and data on few nodes
	CUB [92]	1. High applicability 2. Good storage efficiency	1. A centralised node 2. Hash consistency issue 3. Increase traffic for query
	Jidar [19]	1. High storage efficiency for UTXO-based cryptocurrencies	1. Low data availability 2. Hash consistency issue 3. Increase traffic for query 4. Low applicability (cryptocurrencies based on UTXO)
	MLDC [39]	1. High applicability 2. Low network overhead 3. Low query costs	1. Complexity of node assignment to storage classes
Redaction-based	**Chameleon hash-based** [12]	1. High flexibility for modification	1. Tremendous computational cost 2. Security issue with the trapdoor key 3. Compromise auditability and decentralisation
	Polynomial-based [15]	1. High flexibility for modification	1. Compromise auditability and decentralisation 2. Tradeoffs between security and computational cost 3. Not suitable for PoW
	RSA-based [34]	1. High flexibility to modify with low computational cost	1. Designed for private blockchains computational cost 2. Compromise auditability and decentralisation
	MOF-BC [25]	1. Keep blockchain hash consistency 2. High applicability	1. A centralised component 2. Security issue with key management
	μchain [67]	1. Provide confidentiality by encrypting inactive transactions	1. Reduce auditability 2. Low storage efficiency 3. Low computational efficiency
	LiTiChain [68]	1. Block removal	1. Only for permissioned blockchains 2. Double spending and forge transactions
Content-based	**MOF-BC** [25]	1. Compress various types of contents for IoT	1. Difficulty in verifying original content
	CoinPrune [54]	1. Snapshot of UTXO	1. Tremendous computational cost 2. Not able to remove content on the UTXO set
	Mimblewimble [65]	1. Snapshot of UTXO 2. Improve privacy	1. Increase computational cost and transaction size
	Mina [11]	1. Constant size of ledger 2. Improve privacy	1. Increase computational cost 2. Low transaction throughput

reducing the number of duplicate copies while consuming minimal network and compute resources with a high data availability guarantee. The rest of this section focuses on replication-based optimisation.

In addition to the blockchain scalability trilemma aforementioned, there are tradeoffs between scalability factors. Reducing redundancy, for instance, increases storage efficiency but tends to reduce network efficiency and data availability. We define the evaluation metrics and then survey and analyse the current replication-based approaches. Table 2 shows the results of this analysis on replication-based optimisations, highlighting the scalability trilemma trade-offs. We will discuss their details in the remainder of this section.

6.1 Using External System

Since these types of optimisations utilise external storage instead of local storage to store transactions or blocks, they can effectively eliminate the need for participant nodes to store the entire ledger. As a result, these approaches can significantly reduce the enormous storage requirements.

6.1.1 IPFS-based. IPFS is a distributed data storage system based on content addressing [44, 101] that can store transactions. Valid transactions that miners collect and verify are stored in their local transaction pool and also uploaded to IPFS. IPFS returns their unique hash value, allowing network participants to access the transactions. Miners embed only the hash values provided by IPFS into a candidate block and calculate a Merkle root and a hash of the new block. They then broadcast the new block after they successfully mine it. When other miners receive the new block, they verify it compared to their candidate block. Despite network delay, most transaction hashes in the new block are identical to the hashes in the local transaction pool of miners, while the rest of

Table 2. The Impacts of Replication-based Optimisations

Optimisation	Security			Decentralisation		Scalability		
	C	I	A	DM	DD	NE	CE	SE
IPFS-based	⇑	–	⇓	–	⇓	⇓	–	⇑
CUB	–	–	⇓	⇓	⇓	⇓	–	⇑
Jidar	–	–	⇓	–	⇓	⇓	⇓	⇑
MLDC	⇑	–	⇓	–	⇓	⇓	–	⇑

– : No effect when applying the solution
⇓ : Negative impact of applying the solution
⇑ : Positive impact of applying the solution

the transactions are downloaded from IPFS with their corresponding hashes. Then, miners verify the new block and transactions before appending the block to their ledger.

IPFS-based storage optimisation models show high storage efficiency, because blocks contain only the hash values provided by IPFS, instead of the entire transaction data [87]. However, these IPFS-based optimisations also significantly increase network overhead, as nodes must access IPFS each time to obtain a transaction. Since objects in IPFS are stored and maintained based on content addressing, caching, and pinning, objects can be concentrated on a few nodes. In terms of security, there is a risk of data loss if an object on IPFS is not available, since IPFS does not guarantee data availability. IPFS also has solutions that enforce confidentiality, such as data encryption or chunking, but using hashes to identify objects provides the same level of integrity to the blockchain [20].

6.2 Without External System

These types of approaches optimise the number of replications. These optimisations can significantly reduce storage space required for participant nodes to store transactions or blocks in local storage.

6.2.1 CUB. **Consensus Unit-based Storage Scheme for Blockchain system (CUB)** [92] proposes a storage optimisation scheme where nodes only store partial data of the blockchain ledger in **Consensus Unit (CU)**, which is a group of nodes and maintains at least one copy of the whole ledger. All participants in a blockchain network have a header of blockchain data while blocks are distributed and assigned to different nodes. Each node can verify blocks received from other nodes using the header information. When nodes need blocks they do not have, they must request the blocks from other nodes, increasing the query cost. CUB defines the **Blocks Assignment Optimisation (BAO)** problem, which determines the optimal assignment of blocks to minimise storage space and the query cost. To address this problem, they use heuristic algorithms that consider various situations, including new block arrivals, node joining, and departing, while relying on a centralised master node to assign blocks.

CUB demonstrates an efficient way to reduce the replication of blockchain data for participants by allowing them to store only a part of the ledger [102]. However, increasing storage efficiency also increases query costs, compromising network efficiency. In addition, CUB is also partially centralised, as the master node optimises the allocation of blocks to each node to reduce query costs. However, due to CUB's assumption that nodes within the CU are trusted within a private or permissioned blockchain, there exists a considerable risk of permanent data loss, potentially leading to consistency issues. CUB's main contribution is to optimally assign blocks to minimise storage space and query costs, but reducing redundancy compromises the degree of decentralisation.

6.2.2 Jidar. **Jigsaw-like Data Reduction (Jidar)** [19] is also a storage-optimised approach for reducing the replication of blockchain data based on Bitcoin. Each node in Jidar only stores

relevant and interesting transactions and block headers of all blocks, so it can verify all relevant transactions without communicating with other nodes. Since with only the block headers and relevant transactions a node cannot verify whether the inputs of new transactions are unspent or not, Jidar adds a field of a bloom filter to the block header to validate if the inputs are unspent. However, there is a possibility of the intrinsic false positive matches of the bloom filter. Thus, nodes have to store the entire data of a new block if a false positive matching occurs when they receive a new block. In addition, Jidar introduces a block window in which each node has to add newly received blocks to resolve short-term crashes of nodes, whereas to address long-term crashes, it is recommended that users run multiple instances to ensure data security.

Jidar aims to reduce storage costs by storing only relevant data on each node [41, 102]. However, this approach leads to increased communication costs, as reducing redundancy can result in higher network overhead. Moreover, the utilisation of bloom filters can introduce additional computational expenses. To access the complete blockchain data, a node must collect data from other nodes, but there is a possibility that some nodes may not respond, necessitating the implementation of incentive mechanisms. Additionally, a decrease in the number of full nodes raises the risk of data loss, threatening hash consistency and data availability. The reduction in redundancy also leads to compromises in the degree of decentralisation. Hence, Jidar needs to optimise the number of full nodes to address these concerns.

6.2.3 MLDC. **Multi-Level Distributed Caching (MLDC)** [40] optimises data replication for data-centric blockchain applications such as IoT that store data on the blockchain by leveraging the data access patterns. MLDC assumes a power-law distribution in which data access depends on age—the recent data gets accessed more while the older data gets retrieved seldom. MLDC has a multi-level distributed cache architecture where each node is assigned to a storage class with its own threshold time to remove data in the local storage by their node availability. Over time, each node removes data that has not been recently accessed from the local storage and just stores a hash of the data. A node that does not store the queried data requests the data to a node in the next higher storage class by using content addressing to randomise the destination nodes. The node forwards the data when it receives it from the higher storage class node to the requester.

MLDC enhances storage efficiency and minimises network overhead and query costs in a decentralised manner by optimising the number of nodes and the threshold time for each storage class. The hierarchical structure of MLDC also enhances confidentiality by limiting access to remote queries. However, the tradeoff for reducing redundancy is potential compromise in the degree of decentralisation. Despite this, MLDC demonstrates balanced performance. Nevertheless, attention should be given to the initial allocation of nodes to storage classes.

7 REDACTION-BASED OPTIMISATIONS

The redaction-based approach reduces the size of blockchain data by deleting or updating the data in the ledger through a mining process; each storage node needs a synchronisation process to update new data. These approaches can be classified into three types:

— Some systems use cryptographic schemes to modify blocks, which provides a high level of flexibility but can be expensive (discussed in Section 7.1).
— Another approach generates modified transactions called meta-transactions to update existing transactions (discussed in Section 7.2).
— Last, pruning methods modify or delete transactions or blocks (discussed in Section 7.3).

Since the mining process is involved in redacting data that has been committed to the ledger, redaction-based approaches are expected to affect mining performance. Therefore, focusing on mining performance, we first define the evaluation metrics and then survey and analyse the

Table 3. The Impacts of Replication-based Optimisations

Optimisation	Security			Decentralisation		Scalability		
	C	I	A	DM	DD	NE	CE	SE
Chameleon hash-based	⇓	⇓	–	⇓	–	⇓	⇓	⇑
Polynomial-based	⇓	⇓	–	⇓	–	–	⇓	⇑
RSA-based	⇓	⇓	–	⇓	–	–	–	⇑
MOF-BC	⇓	⇓	–	⇓	–	–	–	⇑
μchain	–	⇓	–	–	–	–	⇓	–
LiTiChain	–	⇓	–	–	–	–	–	⇑

– : No effect when applying the solution
⇓ : Negative impact of applying the solution
⇑ : Positive impact of applying the solution

current redaction-based approaches. Table 3 shows the results of this analysis on replication-based optimisations, highlighting the scalability trilemma trade-offs. We will discuss their details in the remainder of this section.

7.1 Cryptography-based

7.1.1 Chameleon Hash-based. Chameleon Hash is a cryptographic hash function with a trapdoor key that can compute hash collisions [12], allowing modifications to blockchain data while maintaining hash consistency. A redactable blockchain based on the chameleon hash, which can rewrite and remove blocks, is proposed in Reference [2]. It uses a chameleon hash function as an inner hash function to chain blocks in the blockchain and add the randomness field into the block header. When a block is redacted, the randomness field is updated by calculating collision with the trapdoor key, which maintains the blockchain's integrity. The trapdoor key can be managed by a centralised auditor or shared among several parties. The only key managers can redact or remove blocks in the blockchain. When the central authority creates a new redaction block, it broadcasts the block to all participants in the network. The nodes that receive the new modified block verify it and update their local ledger.

Approaches utilising the chameleon hash offer flexibility for modifying, deleting, or inserting, resulting in high storage efficiency. However, these approaches require significant computational costs for calculating hash collisions. Furthermore, the use of a centralised key management system for the trapdoor key can potentially give rise to confidentiality and integrity concerns due to key exposure [99]. Additionally, the complex and costly redacting scheme employed can compromise the efficiency of the blockchain network.

7.1.2 Polynomial-based. Reference [15] proposes a polynomial-based redactable blockchain structure aimed at removing fraudulent transactions. The key concept relies on the utilisation of Lagrange interpolation to construct a unique polynomial of N-1 degree, given N coordinates. Specifically, a unique polynomial for each block is generated using label fields in block header and block body, which represent coordinates. The block body, containing multiple transactions, is organised using the Lagrange interpolation method. Each transaction and a block header are converted into coordinates by a strings-to-integers mapping algorithm called hash-cut mapping. The polynomial function for each block represents its block address, and each block connects to its previous block using the polynomial function recorded in its block header. Additionally, modification difficulty can adjust the degree of computational cost to modify a block to encourage users to accurately record data and reduce the frequency of modification operations. To modify blocks, users are required to recalculate the coordinates that satisfy the unique polynomial and the associated difficulty for each block.

The polynomial-based approach enables redaction and deletion of blocks, and the adjustable modification difficulty offers flexibility to various security requirements and scenarios. However, this approach can compromise auditability, since the modification history is removed. Additionally, high modification difficulty can result in significant modification costs while low modification difficulty can compromise security. In the context of the PoW consensus mechanism, this approach is not well-suited, as modification can lead to a high rate of forks [95]. Furthermore, managing modification permissions in this approach necessitates additional security features such as authentication or supervision. However, these additional security features can potentially undermine decentralisation, which is a fundamental principle of blockchain systems.

7.1.3 RSA-based. Reference [34] presents a redactable blockchain method specifically designed for private blockchains, based on the computational hardness of the **Rivest–Shamir–Adleman (RSA)** problem. Each block consists of three components: a prefix P_i, the actual content C_i, and a suffix X_i. To create a block B_i, a miner randomly selects a value for X_i ranging from 1 and n-1, ensuring that X_i^2 is not 1 (mod n). Subsequently, a hash value h_i is computed by applying a hash function with the concatenation of P_i and C_i where $h_i = HASH(P_i, C_i)$. The resulting hash h_i is then converted into an integer d_i (mod n). The prefix P_{i+1} of the next block is derived using the equation $P_{i+1} = (X_i)^{d_i}$ (mod n). To modify a block B_i, the central authority, possessing the private key, computes a new hash value $h_i' = HASH(P_i, C_i')$, where C_i' is the updated content. The new hash h_i' is then converted into an integer d_i' (mod n), ensuring that d_i' is relatively prime to $\phi(n)$, where $\phi(n)$ represents the Euler function of n. Once d_i' is determined, the central authority can compute the inverse e_i' of d_i' (mod $\phi(n)$) and calculate $X_i' = P_{i+1}^{e_i'}$ (mod n). According to RSA scheme, the integrity of the updated block can be verified through the equation $(X_i')^{d_i'} = (P_{i+1}^{e_i'})^{d_i'} = P_{i+1}$ (mod n).

The RSA-based approach provides a redaction method with robust corruption resistance, utilising RSA scheme at a low computational cost. However, this approach is specifically designed for private blockchains, where a central authority is required to execute the redaction process. Consequently, the involvement of a central authority introduces a centralisation. Furthermore, the removal of modification history can compromise the auditability of the blockchain [95]. For example, malicious attackers can easily revert a block to its previous block, since all versions can successfully pass the validation.

7.2 Meta-transaction

7.2.1 MOF-BC. **Memory Optimised and Flexible Blockchain (MOF-BC)** [25] proposed a flexible, memory-optimised method for IoT users and service providers to remove, summarise, and age transactions in the blockchain. The conventional blockchain does not allow the removal or redaction of data from the blockchain ledger due to its immutable nature. To address this challenge, the hash of blocks in MOF-BC is constructed by hashing the transaction hashes instead of the transaction content, which allows transaction content to be removed while the hash remains stored in the blockchain. A transaction generator can remove its transactions by issuing a remove transaction that indicates transactions to be removed. Miners verify the hashes and the key in the remove transaction to have permission to remove the transactions using the **Generator Verifier (GV)**. In addition, MOF-BC incentivises users to delete, summarise, or age their data according to the reward mechanism. MOF-BC uses a blackboard, a shared read-only central database, to manage the removal history and reward information. While the blackboard can be replicated across multiple nodes for decentralisation, it still results in partial centralisation of MOF-BC. MOF-BC also introduces Summarise and Aging methods, which are optimisations based on content type. We will discuss these methods in Section 8.

By issuing meta-transactions, MOF-BC proposes various methods to reduce the size of blockchain data. It emphasises maintaining hash consistency during transaction redaction. However, the verification process is vulnerable, creating a potential risk of double-spending that can undermine confidentiality and integrity [99]. Additionally, the utilisation of a blackboard approach compromises decentralisation.

7.2.2 μchain. μchain [67] provides a way to modify transactions to handle vulnerable smart contracts or remove abusive content from the blockchain. μchain maintains alternate versions of data records represented by transaction sets containing possible transaction versions to update transactions. There is only one active transaction in a transaction set, while the rest remain inactive alternatives. A set of transactions can be extended later by issuing extended transactions referencing their mutable transaction. To update an active transaction, a mutant transaction that defines a new active transaction is issued by legitimate users specified in the transaction policy. Validators check that the transaction policy has been met in the process of mining. Mutant transactions can be issued multiple times, so μchain proposes a transaction tree structure for managing transaction history. To solve the confidentiality problems, μchain encrypts the transactions in mutable transaction sets with transaction-specific keys. The keys are managed by **Encryption with Secret Sharing (ESS)**. The transaction-specific keys are split into shares and distributed among validators. The keys can only be reconstructed when enough shares have been collected.

While μchain permits transaction modification, it does not delete inactive transactions from the blockchain record; instead, it encrypts them to conceal their visibility. As a result, there is no improvement in storage efficiency. Furthermore, encrypting transactions necessitates additional computational resources and undermines auditability [66]. The use of encryption to enhance confidentiality also raises the risk of key exposure, potentially compromising integrity.

7.3 Pruning-based

7.3.1 LiTiChain. **Blockchain of Finite-Lifetime Blocks (LiTiChain)** [68] aims to provide a way to tackle the storage issue for IoT edge nodes by removing expired blocks. The immutable and ever-growing nature of blockchain usually leads to a lack of storage in IoT edge servers with limited storage space. To address this issue, transactions and blocks in LiTiChain have a finite lifetime, so the expired block can be removed from the chain while maintaining its connectivity. Blocks in conventional blockchains are linked with previous blocks in the order of creation. In contrast, blocks in LiTiChain are linked with other blocks in the tree structure, not only in the order of creation but also in the order of expiration. If a block is followed by another block that has not yet expired, then it cannot be removed even if it expires. It must remain in the chain to keep the consistency of the chain until the following block expires. In addition, an indefinite type of transaction whose lifetime is not determined at creation time can be renewed by creating a new block containing the indefinite transaction and then the old block can be deleted.

LiTiChain provides a solution for deleting blocks from the blockchain, making it suitable for resource-constrained edge-based IoT systems [29, 93]. The block removal feature enables high storage efficiency. However, it is important to note that LiTiChain is specifically designed for permissioned blockchains and lacks provisions for ensuring confidentiality. Additionally, there are no limitations on transaction expiry and reissuance, which leaves room for malicious nodes to issue forged or double-spending transactions, posing a potential threat to integrity.

8 CONTENT-BASED OPTIMISATIONS

Content-based approaches reduce the size of blockchain data itself by compressing the data based on the content type before or after the data is mined. Most of these approaches focus on reducing

Table 4. The Impacts of Replication-based Optimisations

Optimisation	Security			Decentralisation		Scalability		
	C	I	A	DM	DD	NE	CE	SE
MOF-BC	⇓	⇓	–	⇓	–	⇑	⇓	⇑
CoinPrune	–	–	⇓	–	⇓	⇑	⇓	⇑
Mimblewimble	⇑	⇑	–	–	–	⇓	⇓	⇑
Mina	⇑	⇑	–	–	–	⇓	⇓	⇑

– : No effect when applying the solution
⇓ : Negative impact of applying the solution
⇑ : Positive impact of applying the solution

the asset transfer history for cryptocurrencies like UTXO (discussed in Section 8.2), while some focus on data-centric blockchain, for example, reducing the size of IoT sensor data (discussed in Section 8.1). Table 4 shows the results of this analysis on content-based optimisations, highlighting the scalability trilemma trade-offs. We will examine the details of these trade-offs for each solution in the following sections.

8.1 Data-centric

8.1.1 MOF-BC. The "Summarise" method of MOF-BC [25] aims to reduce the transaction history of asset transfers, specifically for transactions similar to Bitcoin. This method involves summarising multiple transactions into a single consolidated transaction by issuing a summary transaction. Miners utilise GV to verify the generator's authority to summarise the listed transactions in the summary transaction. However, the "aging" method of MOF-BC provides a way to compress the stored data based on the content type, specifically for IoT blockchain. In this method, the user, who is the generator of the original transaction, can compress the data by using various compression ways including lossy or lossless compression [58]. The user then broadcasts the aged transaction containing the compressed data and miners verify the transaction by checking the permission to modify the original transaction. The miners add the aged transaction into a new block as a regular transaction and broadcast it. When storage nodes receive the aged transaction in the new block, they can remove the corresponding original transaction from the blockchain ledger. MOF-BC also uses the aforementioned blackboard to manage this revision history.

According to the content type, MOF-BC can compress stored data in the ledger to improve storage efficiency and network efficiency, whereas compressing contents negatively impacts computational efficiency. In particular, lossy compression, which provides high compression [48]. However, when the content is compressed by lossy compression, it is difficult to verify whether the aged transaction is generated from its original transaction. Additionally, MOF-BC utilises a blackboard approach, which is a centralised component.

8.2 Cryptocurrency

8.2.1 CoinPrune. CoinPrune [54] proposes a snapshot-based pruning scheme to deal with storage and performance issues of public blockchain by taking a snapshot of ***Unspent Transaction Outputs* (UTXO)**. Miners in CoinPrune take a snapshot of their current set of UTXO in constant intervals, which allows new nodes to join with only partial data including block headers and the snapshot instead of downloading entire historical blockchain data. Pulse blocks are periodically issued (e.g., every 10,000 blocks) to synchronise the creation of a snapshot by miners in CoinPrune. A new snapshot is then generated as soon as a pulse block has been mined. After the new snapshot is created, miners reaffirm the snapshot by adding the identifier of the snapshot into the coinbase transaction in their mining block until the reaffirmation window expires. These reaffirmation

activities prevent malicious nodes from deploying incorrect snapshots. A new joining node only needs to download the current snapshot and some data from the ledger such as headerchain, which is a chain of blockheaders, and chaintail, which is the remaining full blocks succeeding the snapshot's block.

CoinPrune reduces synchronisation time for joining nodes by creating snapshots of the Bitcoin UTXO set, which improves network efficiency. However, the process of taking a snapshot incurs a significant computational burden. Additionally, CoinPrune does not actually delete data from the blockchain; instead, certain nodes manage the snapshot, similar to the lightweight nodes in Bitcoin. It remains essential to have full nodes storing the entire ledger to manage the system. Therefore, reducing the number of full nodes can compromise the degree of decentralisation.

8.2.2 Mimblewimble. Mimblewimble [65] is a blockchain-based cryptocurrency designed to compact the transaction history with simple verification and to achieve strong user privacy by using confidential transactions. A Mimblewimble transaction consists of inputs, outputs with range proofs, and an excess with a summable sinking signature calculated by a homomorphic commitment scheme. Since the homomorphic commitment scheme encrypts inputs and outputs, a sender and receiver participate in the transaction to share the decryption key. There are two types of inputs. Explicit inputs have no reference to an output of a previous transaction, while non-explicit inputs have a reference to an output of a transaction in an earlier block. From the genesis block to the latest block, non-explicit inputs and their referenced output can be deleted as spent money. Therefore, Mimblewimble can combine transactions from multiple blocks into a single block.

Mimblewimble can remove information of unnecessary transactions by combining multiple transactions, which reduces blockchain data size. However, the homomorphic commitment scheme requires more computational costs and increases the size of transaction. In addition, the increase in the transaction size compromises network efficiency. However, in terms of security, consolidating multiple transactions into a single block by a homomorphic commitment scheme hides information, so it fully provides confidentiality and integrity of each transaction. Consequently, combining multiple blocks into a single block can reduce the ledger size by eliminating unnecessary transaction information.

8.2.3 Mina. Mina [11] is an account-based blockchain that aims to provide efficient verification and a small ledger size by using a **succinct non-interactive argument of knowledge (SNARK)** [5]. Unlike the UTXO-based models, the current state of blockchain in Mina is the aggregated state of all accounts such as Ethereum, and the state can be changed by committing transactions called witnesses. A blockchain summary consists of the blockchain state and the proof, and a new blockchain summary can be computed from the current blockchain summary and a new block by the chain summary update algorithm. In Mina, full nodes only need the blockchain summary to verify blockchain correctness. Block producers generate a block, while blockchain summary producers generate blockchain summaries. To manage new states at a constant size, Mina proposes a PoS consensus protocol for succinct blockchains called Ouroboros Samasika, where an incrementally computable SNARK verifies the state.

Mina can significantly reduce the size of blockchain data by using its own consensus protocol and the incrementally computable SNARK. The blockchain summary using SNARK also enhances integrity and confidentiality by hiding information, while the complexity of SNARK greatly reduces computational efficiency. In addition, verifying each transaction using SNARK decreases network efficiency.

Table 5. Optimising Processes

Optimisations	Pre-processing	Mining	Post-processing
Replication-based optimisation			
IPFS-based [44, 101]	Upload transactions to IPFS and receive their hash, Generate a transaction with the hashes	Verify transactions and commit them	Read transactions from IPFS
CUB [92]			Optimise block assignment in a CU
Jidar [19]		Maintain UTXO with Bloom filter	Nodes store only relevant transactions and the block headers
MLDC [39]			Nodes remove not accessed data in the local storage and only store a hash of the data
Redaction-based optimisation			
Chameleon hash-based [12]		Find hash collision of the chameleon hash with the trapdoor key	
Polynomial-based [15]		Compute a new integer and its inverse corresponding to a hash value for the updated message	
RSA-based [34]		Compute a new coordinate for the updated message	
MOF-BC [25]	Generate a remove transaction and broadcast it	Verify a remove transaction and commit it	Nodes remove the old transaction from the local storage
μchain [67]	Generate a mutable or mutant transaction	Verify a mutable and mutant transaction and commit it	
LiTiChain [68]	Generate a transaction with finite lifetimes	Add the lifetime of a block and renew a block or delete an expired block	Nodes remove expired blocks from the local storage
Content-based optimisation			
MOF-BC [25]	Compress data and generate a summary or aged transaction	Verify an aged transaction and commit it	Nodes delete the old transaction
CoinPrune [54]	Periodically issue pulse blocks	Create a UTXO snapshot	
Mimblewimble [65]	Generate confidential transactions	Aggregate all transactions and verify them, Compute a sinking signature, Remove spent inputs and outputs and generate a new block	
Mina [11]		Randomly select miners and compute SNARK proofs, Produce a new state of the blockchain	

9 COMPARISONS

To overcome the scalability trilemma, many blockchain storage optimisations strive to minimise side effects in various ways. Table 5 shows optimisation processes at the steps of the blockchain data management process. Replication-based optimisations mainly focus on the post-processing to optimise the number of replications. As these approaches tend to minimise modifications to the

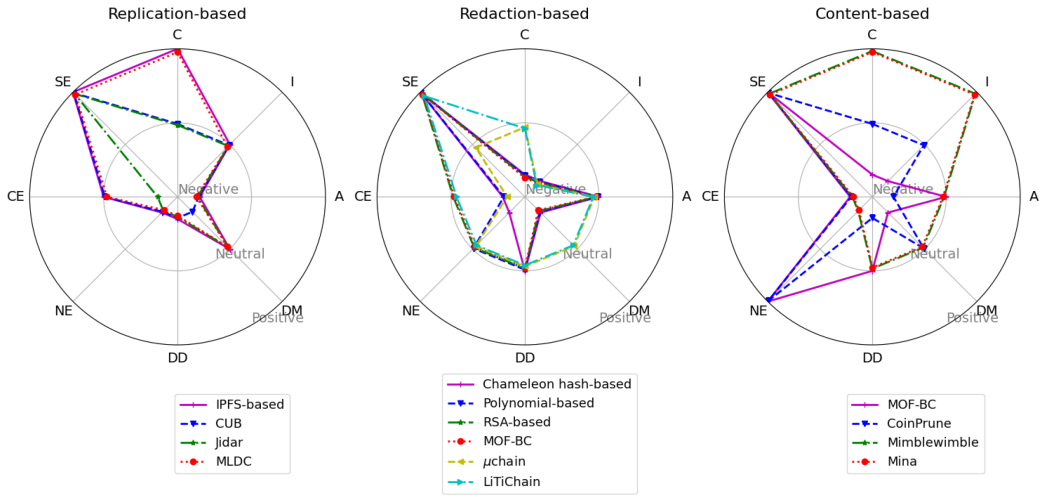


Fig. 6. Comparison of different optimisation schemes.

blockchain protocol, most of them can be easily applicable to any blockchain system. In contrast, the majority of optimisation processes for the redaction-based solutions takes place through the entire phase, issuing a new version of data, committing it, and deleting the old version. Most of them redefine the structure of transactions or blocks, while some of them require additional components to manage their optimisation scheme. In the case of content-based solutions, there are two types of optimisations: one is to compress contents before mining, while the other is to compress contests after mining. They collect the same type of contents, compress them with their own solution, and then commit the compressed content or update it with the compressed version.

As shown in Figure 6, the scalability factors are measured based on the evaluation metrics defined in previous sections along with security and decentralisation.

— *In general, replication-based optimisations often involve a tradeoff, as these solutions aim to improve storage efficiency but compromise data availability, the degree of decentralisation, and network efficiency. This compromise arises due to the reducing redundancy, which increases remote queries. While it is inevitable to make compromises in data availability, decentralisation and network efficiency, each solution varies in the extent of these compromises, striving to minimise degradation in different ways. Compared to other optimisations, replication-based solutions offer relatively high adaptability, since these solutions employ simple mechanisms to handle replication data.*

— *Redaction-based solutions generally raise security concerns due to their capability to delete or modify information that is already committed on the blockchain ledger. Consequently, the integrity of the ledger can be compromised by such actions. Furthermore, the introduction of optimisation processes to redact data in the ledger often leads to a compromise in computational efficiency, while network efficiency remains largely unaffected. As for decentralisation, some optimisations incorporate centralised components for system management.*

— *Finally, compressing content necessitates significant computational costs but effectively enhances storage efficiency. Most content-based optimisations for cryptocurrency employ cryptographic algorithms to compress the history of transactions, which requires expensive computations. As a consequence, these optimisations offer good security but poor computational and network efficiency. While the reduced data size improves storage efficiency, the expensive*

computations and complex management processes decrease computational efficiency and transaction throughput.

10 OPEN OPPORTUNITIES AND CHALLENGES

Blockchain storage optimisations have drawn significant attention due to the ever-increasing ledger size and demand for data redaction. However, research on blockchain storage optimisation is still in its infancy. We summarise the open opportunities and challenges below based on our survey.

10.1 On the Effectiveness of Replication-based Optimisation

Most replication-based optimisations do not require changes to the consensus protocol, or they only require simple changes. They can operate at the storage layer independently of the blockchain. Hence, these solutions are a broadly applicable, platform-agnostic solution to existing blockchains. Most solutions in this category propose efficient ways to reduce data replication while minimising query costs and network overhead. However, reducing redundancy can also compromise data availability and network efficiency. Given these, there is a research opportunity to investigate the effect of lowering redundancy on data availability and network efficiency. The specific question is: "*How to optimise the performance of replication-based storage approaches without compromising availability and network efficiency?*"

10.2 Limitations of Redaction-based Optimisation

The redaction-based solutions aim to remove data from the blockchain ledger. It is important to note that immutability is one of the fundamental properties of blockchain technology. Allowing the stakeholders to remove data from the ledger is against the spirit of the blockchain. However, the redaction-based storage optimisations focus on preserving the integrity of the blockchain despite removing transactions from the blockchain through innovative cryptographic schemes such as chameleon hash, polynomial, or RSA. At the same time, many redaction-based solutions suffer from redaction history management and cryptographic key management. Some solutions follow a centralised architecture, while others break blockchain hash consistency. These problems highlight the need to build protocols to redact data following a decentralised architecture with transparency and auditability. The specific research question that emerges is: "*How to remove transactions from the blockchain while guaranteeing integrity, transparency, and decentralisation?*"

10.3 Generality of Content-based Solutions

Content-based solutions compress the transaction data, minimising the storage cost. Most of the existing solutions in this category apply to cryptocurrency applications. However, several non-cryptocurrency applications such as IoT [24, 26], supply chains [50, 71], and data marketplaces [69, 72, 91] employ blockchain for enhancing trust in multi-stakeholder applications. For instance, due to the various uses of IoT, IoT data collected from sensors can be of many types, from integer to text or audio to video streams [49]. To reduce the data size of content, various mechanisms can be applied, depending on the content types, such as lossy or lossless compression [48, 79], Dimensionality reduction [13, 73], or Deduplication [38, 51]. These mechanisms highlight the wide array of content-based solutions. The specific research question is therefore: "*How to optimise content for blockchain-based applications in an application-agnostic fashion?*"

10.4 Hybrid Optimisation Techniques

Existing replication-based, redaction-based, and content-based optimisation approaches offer various benefits to the blockchain while introducing some limitations. Given the benefits of these

approaches, there is an opportunity to combine one or more of these approaches to develop a more robust storage optimisation solution, offering high efficiency while minimising compromise on data availability and decentralisation. For example, replication-based optimisation can be easily combined with redaction-based or content-based optimisations. Therefore, future research on blockchain storage optimisation may explore combinations of multiple optimisation approaches at various layers. The specific research question is: "*How to develop a novel storage optimisation solution by combining existing and proven optimisation techniques?*"

10.5 Lack of Holistic and Sustainable Approaches

As shown in Figure 5, blockchain platforms such as Bitcoin and Ethereum tweak the block size and the number of transactions in the block to increase the throughput demands or enhance the scalability. Such approaches seem to give less importance to storage, as increasing the number of blocks or transactions would maximise the storage cost. The research community must develop a systematic and holistic approach to ensure that future changes to the blockchain platforms balance the storage, network, and computational efficiency to sustain the growth and adoption of blockchains. The specific research question is "*How to upgrade blockchains following a holistic and sustainable framework?*"

11 CONCLUSION

The storage requirement of blockchains is increasing rapidly, demanding solutions to optimise the storage cost. We surveyed the state-of-the-art blockchain storage optimisation solutions and classified the existing solutions into replication-based, redaction-based, and content-based optimisations. We have also dissected the current solutions to explain the operations executed at different layers of the blockchain for each optimisation solution. Existing solutions are compared and contrasted to highlight their security, decentralisation, and scalability properties. We have shown that the replication-based optimisations reduce the duplication of committed data in the blockchain ledger. Redaction-based optimisations modify committed data by replacing it with a new version, altering the blockchain without compromising its integrity. Content-based optimisations mainly compress data before committing it. Our analysis has helped us expose several limitations and opportunities for future research, which we believe is invaluable to researchers and practitioners in the blockchain domain.

REFERENCES

[1] Amani Altarawneh, Tom Herschberg, Sai Medury, Farah Kandah, and Anthony Skjellum. 2020. Buterin's scalability trilemma viewed through a state-change-based classification for common consensus algorithms. In *10th Annual Computing and Communication Workshop and Conference (CCWC'20)*. 0727–0736. DOI: https://doi.org/10.1109/CCWC47524.2020.9031204

[2] Giuseppe Ateniese, Bernardo Magri, Daniele Venturi, and Ewerton Andrade. 2017. Redactable blockchain–or–rewriting history in Bitcoin and friends. In *IEEE European Symposium on Security and Privacy (EuroS&P'17)*. IEEE, 111–126.

[3] Fadi Barbàra and Claudio Schifanella. 2022. BxTB: Cross-chain exchanges of Bitcoins for all Bitcoin wrapped tokens. In *4th International Conference on Blockchain Computing and Applications (BCCA'22)*. IEEE, 143–150.

[4] Alireza Beikverdi and JooSeok Song. 2015. Trend of centralization in Bitcoin's distributed network. In *IEEE/ACIS 16th International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD'15)*. IEEE, 1–6.

[5] Eli Ben-Sasson, Alessandro Chiesa, Eran Tromer, and Madars Virza. 2014. Succinct Non-interactive zero knowledge for a Von Neumann architecture. In *23rd USENIX Security Symposium (USENIX Security'14)*. 781–796.

[6] Juan Benet. 2014. IPFS - Content Addressed, Versioned, P2P File System. arXiv:1407.3561 [cs.NI].

[7] J. Benet and N. Greco. 2017. *Filecoin: A Decentralized Storage Network*. Whitepaper. Protocol Labs. Retrieved from https://filecoin.io/filecoin.pdf

[8] Nazanin Zahed Benisi, Mehdi Aminian, and Bahman Javadi. 2020. Blockchain-based decentralized storage networks: A survey. *J. Netw. Comput. Applic.* 162 (2020), 102656.
[9] Mirko Bez, Giacomo Fornari, and Tullio Vardanega. 2019. The scalability challenge of Ethereum: An initial quantitative analysis. In *IEEE International Conference on Service-Oriented System Engineering (SOSE'19)*. IEEE, 167–176.
[10] Blockchair. 2022. Blockchain explorer, analytics and web services. Retrieved from https://blockchair.com/bitcoin
[11] Joseph Bonneau, Izaak Meckler, Vanishree Rao, and Evan Shapiro. 2020. *Mina: Decentralized Cryptocurrency at Scale*. Whitepaper New York University O (1) Labs, New York, NY, 1–47.
[12] Jan Camenisch, David Derler, Stephan Krenn, Henrich C. Pöhls, Kai Samelin, and Daniel Slamanig. 2017. Chameleon-hashes with ephemeral trapdoors. In *IACR International Workshop on Public Key Cryptography*. Springer, 152–182.
[13] S. Kalyan Chakravarthy, N. Sudhakar, E. Srinivasa Reddy, D. Venkata Subramanian, and P. Shankar. 2019. Dimension reduction and storage optimization techniques for distributed and big data cluster environment. In *Soft Computing and Medical Bioinformatics*. Springer, 47–54.
[14] Huan Chen and Yijie Wang. 2020. MiniChain: A lightweight protocol to combat the UTXO growth in public blockchain. *J. Parallel Distrib. Comput.* 143 (2020), 67–76.
[15] Lichen Cheng, Jiqiang Liu, Chunhua Su, Kaitai Liang, Guangquan Xu, and Wei Wang. 2019. Polynomial-based modifiable blockchain structure for removing fraud transactions. *Fut. Gen. Comput. Syst.* 99 (2019), 154–163.
[16] Shaen Corbet, Brian Lucey, and Larisa Yarovaya. 2018. Datestamping the Bitcoin and Ethereum bubbles. *Finan. Res. Lett.* 26 (2018), 81–88.
[17] Kyle Croman, Christian Decker, Ittay Eyal, Adem Efe Gencer, Ari Juels, Ahmed Kosba, Andrew Miller, Prateek Saxena, Elaine Shi, Emin Gün Sirer, Dawn Song, and Roger Wattenhofer. 2016. On scaling decentralized blockchains: (A Position Paper). In *International Conference on Financial Cryptography and Data Security*, Springer, 106–125.
[18] Hong-Ning Dai, Zibin Zheng, and Yan Zhang. 2019. Blockchain for internet of things: A survey. *IEEE Internet Things J.* 6, 5 (2019), 8076–8094.
[19] Xiaohai Dai, Jiang Xiao, Wenhui Yang, Chaofan Wang, and Hai Jin. 2019. Jidar: A jigsaw-like data reduction approach without trust assumptions for Bitcoin system. In *IEEE 39th International Conference on Distributed Computing Systems (ICDCS'19)*. IEEE, 1317–1326.
[20] Erik Daniel and Florian Tschorsch. 2022. IPFS and friends: A qualitative comparison of next generation peer-to-peer data networks. *IEEE Commun. Surv. Tutor.* 24, 1 (2022), 31–52.
[21] Sergi Delgado-Segura, Cristina Pérez-Sola, Guillermo Navarro-Arribas, and Jordi Herrera-Joancomartí. 2019. Analysis of the Bitcoin UTXO set. In *International Workshops on Financial Cryptography and Data Security (FC'18)* . Springer, 78–91.
[22] Sergi Delgado-Segura, Cristina Pérez-Solà, Guillermo Navarro-Arribas, and Jordi Herrera-Joancomartí. 2020. A fair protocol for data trading based on Bitcoin transactions. *Fut. Gen. Comput. Syst.* 107 (2020), 832–840.
[23] Bassirou Diène, Ousmane Diallo, Joel J. P. C. Rodrigues, E. L. Hadji M. Ndoye, and Ciprian Teodorov. 2020. Data management mechanisms for IoT: Architecture, challenges and solutions. In *5th International Conference on Smart and Sustainable Technologies (SpliTech'20)*. IEEE, 1–6.
[24] Ali Dorri, Salil S. Kanhere, and Raja Jurdak. 2017. Towards an optimized blockchain for IoT. In *IEEE/ACM 2nd International Conference on Internet-of-Things Design and Implementation (IoTDI'17)*. IEEE, 173–178.
[25] Ali Dorri, Salil S. Kanhere, and Raja Jurdak. 2019. MOF-BC: A memory optimized and flexible blockchain for large scale networks. *Fut. Gen. Comput. Syst.* 92 (2019), 357–373.
[26] Ali Dorri, Salil S. Kanhere, Raja Jurdak, and Praveen Gauravaram. 2017. Blockchain for IoT security and privacy: The case study of a smart home. In *IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops'17)*. IEEE, 618–623.
[27] Martin Florian, Sebastian Henningsen, Sophie Beaucamp, and Björn Scheuermann. 2019. Erasing data from blockchain nodes. In *IEEE European Symposium on Security and Privacy Workshops (EuroS&PW'19)*. IEEE, 367–376.
[28] Daniel Fullmer and A. Stephen Morse. 2018. Analysis of difficulty control in Bitcoin and proof-of-work blockchains. In *IEEE Conference on Decision and Control (CDC'18)*. IEEE, 5988–5992.
[29] Thippa Reddy Gadekallu, Quoc-Viet Pham, Dinh C. Nguyen, Praveen Kumar Reddy Maddikunta, Natarajan Deepa, B. Prabadevi, Pubudu N. Pathirana, Jun Zhao, and Won-Joo Hwang. 2021. Blockchain for edge of things: Applications, opportunities, and challenges. *IEEE Internet Things J.* 9, 2 (2021), 964–988.
[30] Karan Singh Garewal. 2020. *Cryptocurrency Transaction Processing*. Apress, Berkeley, CA, 113–136. https://doi.org/10.1007/978-1-4842-5893-4_9
[31] Victor Gayoso Martinez, Luis Hernández-Álvarez, and Luis Hernandez Encinas. 2020. Analysis of the cryptographic tools for blockchain and Bitcoin. *Mathematics* 8, 1 (2020), 131.
[32] Suman Ghimire and Henry Selvaraj. 2018. A survey on Bitcoin cryptocurrency and its mining. In *26th International Conference on Systems Engineering (ICSEng'18)*. IEEE, 1–6.

[33] Toby Gibbs and Suwaree Yordchim. 2014. Thai perception on Litecoin value. *Int. J. Soc., Behav., Educ., Econ., Busin. Industr. Eng.* 8, 8 (2014), 2613–5.
[34] Dima Grigoriev and Vladimir Shpilrain. 2021. RSA and redactable blockchains. *Int. J. Comput. Math.: Comput. Syst. Theor.* 6, 1 (2021), 1–6.
[35] Martin Haferkorn and Josué Manuel Quintana Diaz. 2015. Seasonality and interconnectivity within cryptocurrencies—An analysis on the basis of Bitcoin, Litecoin and Namecoin. In *7th International Workshop on Enterprise Applications and Services in the Finance Industry (FinanceCom'14)*. Springer, 106–120.
[36] Abdelatif Hafid, Abdelhakim Senhaji Hafid, and Mustapha Samih. 2020. Scaling blockchains: A comprehensive survey. *IEEE Access* 8 (2020), 125244–125262.
[37] A. K. M. Bahalul Haque, A. K. M. Najmul Islam, Sami Hyrynsalmi, Bilal Naqvi, and Kari Smolander. 2021. GDPR compliant blockchains—A systematic literature review. *IEEE Access* 9 (2021), 50593–50606.
[38] Qinlu He, Zhanhuai Li, and Xiao Zhang. 2010. Data deduplication techniques. In *International Conference on Future Information Technology and Management Engineering*, Vol. 1. IEEE, 430–433.
[39] Jun Wook Heo, Ali Dorri, and Raja Jurdak. 2022. Multi-level distributed caching on the blockchain for storage optimisation. In *IEEE International Conference on Blockchain and Cryptocurrency (ICBC'22)*. Institute of Electrical and Electronics Engineers Inc.
[40] Jun Wook Heo, Gowri Sankar Ramachandran, Ali Dorri, and Raja Jurdak. 2022. Blockchain storage optimisation with multi-level distributed caching. *IEEE Trans. Netw. Serv. Manag.* 19, 4 (2022), 3724–3736.
[41] Huawei Huang, Wei Kong, Sicong Zhou, Zibin Zheng, and Song Guo. 2021. A survey of state-of-the-art on blockchains: Theories, modelings, and tools. *ACM Comput. Surv.* 54, 2 (2021), 1–42.
[42] Ke Huang, Xiaosong Zhang, Yi Mu, Fatemeh Rezaeibagha, and Xiaojiang Du. 2021. Scalable and redactable blockchain with update and anonymity. *Inf. Sci.* 546 (2021), 25–41.
[43] Puneet Kumar Kaushal, Amandeep Bagga, and Rajeev Sobti. 2017. Evolution of Bitcoin and security risk in Bitcoin wallets. In *International Conference on Computer, Communications and Electronics (Comptelix'17)*. IEEE, 172–177.
[44] Randhir Kumar and Rakesh Tripathi. 2019. Implementation of distributed file storage and access framework using IPFS and blockchain. In *5th International Conference on Image Information Processing (ICIIP'19)*. IEEE, 246–251.
[45] Kejiao Li, Hui Li, Hanxu Hou, Kedan Li, and Yongle Chen. 2017. Proof of vote: A high-performance consensus protocol based on vote mechanism & consortium blockchain. In *IEEE 19th International Conference on High Performance Computing and Communications; IEEE 15th International Conference on Smart City; IEEE 3rd International Conference on Data Science and Systems (HPCC/SmartCity/DSS'17)*. IEEE, 466–473.
[46] Haojun Liu, Xinbo Luo, Hongrui Liu, and Xubo Xia. 2021. Merkle tree: A fundamental component of blockchains. In *International Conference on Electronic Information Engineering and Computer Science (EIECS'21)*. IEEE, 556–561.
[47] Yulin Liu, Luyao Zhang, and Yinhong Zhao. 2022. Deciphering Bitcoin blockchain data by cohort analysis. *Scient. Data* 9, 1 (2022), 136.
[48] Shaofei Lu, Qinhua Xia, Xiaolin Tang, Xuyang Zhang, Yingping Lu, and Jingke She. 2021. A reliable data compression scheme in sensor-cloud systems based on edge computing. *IEEE Access* 9 (2021), 49007–49015.
[49] Meng Ma, Ping Wang, and Chao-Hsien Chu. 2013. Data management for internet of things: Challenges, approaches and opportunities. In *IEEE International Conference on Green Computing and Communications and IEEE Internet of Things and IEEE Cyber, Physical and Social Computing*. IEEE, 1144–1151.
[50] Sidra Malik, Salil S. Kanhere, and Raja Jurdak. 2018. ProductChain: Scalable blockchain framework to support provenance in supply chains. In *IEEE 17th International Symposium on Network Computing and Applications (NCA'18)*. IEEE, 1–10.
[51] E. Manogar and S. Abirami. 2014. A study on data deduplication techniques for optimized storage. In *6th International Conference on Advanced Computing (ICoAC'14)*. IEEE, 161–166.
[52] Roman Matzutt, Martin Henze, Jan Henrik Ziegeldorf, Jens Hiller, and Klaus Wehrle. 2018. Thwarting unwanted blockchain content insertion. In *IEEE International Conference on Cloud Engineering (IC2E'18)*. IEEE, 364–370.
[53] Roman Matzutt, Jens Hiller, Martin Henze, Jan Henrik Ziegeldorf, Dirk Müllmann, Oliver Hohlfeld, and Klaus Wehrle. 2018. A quantitative analysis of the impact of arbitrary blockchain content on Bitcoin. In *International Conference on Financial Cryptography and Data Security*. Springer, 420–438.
[54] Roman Matzutt, Benedikt Kalde, Jan Pennekamp, Arthur Drichel, Martin Henze, and Klaus Wehrle. 2020. How to securely prune Bitcoin's blockchain. In *IFIP Networking Conference (Networking'20)*. IEEE, 298–306.
[55] Du Mingxiao, Ma Xiaofeng, Zhang Zhe, Wang Xiangwei, and Chen Qijun. 2017. A review on consensus algorithm of blockchain. In *IEEE International Conference on Systems, Man, and Cybernetics (SMC'17)*. IEEE, 2567–2572.
[56] Satoshi Nakamoto. 2009. Bitcoin: A peer-to-peer electronic cash system. Retrieved from https://bitcoin.org/en/bitcoin-paper
[57] Muhammad Hassan Nasir, Junaid Arshad, Muhammad Mubashir Khan, Mahawish Fatima, Khaled Salah, and Raja Jayaraman. 2022. Scalable blockchains—A systematic review. *Fut. Gen. Comput. Syst.* 126 (2022), 136–162.

[58] Suman Nath. 2009. Energy efficient sensor data logging with amnesic flash storage. In *International Conference on Information Processing in Sensor Networks*. IEEE, 157–168.
[59] Pranav Nerurkar, Dhiren Patel, Yann Busnel, Romaric Ludinard, Saru Kumari, and Muhammad Khurram Khan. 2021. Dissecting Bitcoin blockchain: Empirical analysis of Bitcoin network (2009–2020). *J. Netw. Comput. Applic.* 177 (2021), 102940.
[60] Sina Rafati Niya, Julius Willems, and Burkhard Stiller. 2021. On-chain IoT data modification in blockchains. *arXiv preprint arXiv:2103.10756* (2021).
[61] Kazım Rıfat Özyılmaz, Harsh Patel, and Ankit Malik. 2018. Split-scale: Scaling Bitcoin by partitioning the UTXO space. In *IEEE 9th International Conference on Software Engineering and Service Science (ICSESS'18)*. IEEE, 41–45.
[62] Russell O'Connor and Marta Piekarska. 2017. Enhancing Bitcoin transactions with covenants. In *International Workshops on Financial Cryptography and Data Security (FC'17)*. Springer, 191–198.
[63] M. Padmavathi and R. M. Suresh. 2019. Secure P2P intelligent network transaction using Litecoin. *Mob. Netw. Applic.* 24 (2019), 318–326.
[64] Huma Pervez, Muhammad Muneeb, Muhammad Usama Irfan, and Irfan Ul Haq. 2018. A comparative analysis of DAG-based blockchain architectures. In *12th International Conference on Open Source Systems and Technologies (ICOSST'18)*. IEEE, 27–34.
[65] Andrew Poelstra. 2016. *Mimblewimble*. White paper (2016). https://download.wpsoftware.net/bitcoin/wizardry/mimblewimble.pdf
[66] Eugenia Politou, Fran Casino, Efthymios Alepis, and Constantinos Patsakis. 2019. Blockchain mutability: Challenges and proposed solutions. *IEEE Trans. Emerg. Topics Comput.* 9, 4 (2019), 1972–1986.
[67] Ivan Puddu, Alexandra Dmitrienko, and Srdjan Capkun. 2017. μchain: How to forget without hard forks. *IACR Cryptology ePrint Archive 2017/106* (2017).
[68] Chan Kyu Pyoung and Seung Jun Baek. 2019. Blockchain of finite-lifetime blocks with applications to edge-based IoT. *IEEE Internet Things J.* 7, 3 (2019), 2102–2116.
[69] Rahul Radhakrishnan, Gowri Sankar Ramachandran, and Bhaskar Krishnamachari. 2019. SDPP: Streaming data payment protocol for data economy. In *IEEE International Conference on Blockchain and Cryptocurrency (ICBC'19)*. IEEE, 17–18.
[70] Gowri Sankar Ramachandran and Bhaskar Krishnamachari. 2018. Blockchain for the IoT: Opportunities and challenges. *arXiv preprint arXiv:1805.02818* (2018).
[71] Gowri Sankar Ramachandran, Sidra Malik, Shantanu Pal, Ali Dorri, Volkan Dedeoglu, Salil Kanhere, and Raja Jurdak. 2021. Blockchain in supply chain: Opportunities and design considerations. *arXiv preprint arXiv:2108.12032* (2021).
[72] Gowri Sankar Ramachandran, Rahul Radhakrishnan, and Bhaskar Krishnamachari. 2018. Towards a decentralized data marketplace for smart cities. In *IEEE International Smart Cities Conference (ISC2'18)*. 1–8. DOI: https://doi.org/10.1109/ISC2.2018.8656952
[73] Ridhima Rani, Meenu Khurana, Deepika Sharma, and Aditi Moudgil. 2021. Comparative study on various storage optimization techniques in IoT-cloud ecosystem. In *International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE'21)*. IEEE, 659–663.
[74] Spyridon Samonas and David Coss. 2014. The CIA strikes back: Redefining confidentiality, integrity and availability in security. *J. Inf. Syst. Secur.* 10, 3 (2014).
[75] Maurice Schellekens. 2019. Does regulation of illegal content need reconsideration in light of blockchains? *Int. J. Law Inf. Technol.* 27, 3 (2019), 292–305.
[76] Simon Schwerin. 2018. Blockchain and privacy protection in the case of the European general data protection regulation (GDPR): A delphi study. *J. Brit. Blockchain Assoc.* 1, 1 (2018).
[77] Pratima Sharma, Rajni Jindal, and Malaya Dutta Borah. 2020. Blockchain technology for cloud storage: A systematic literature review. *ACM Comput. Surv.* 53, 4 (2020), 1–32.
[78] Wellington Fernandes Silvano and Roderval Marcelino. 2020. IOTA tangle: A cryptocurrency to communicate internet-of-things data. *Fut. Gen. Comput. Syst.* 112 (2020), 307–319.
[79] Pallavi Srivastava and Navish Garg. 2015. Secure and optimized data storage for IoT through cloud framework. In *International Conference on Computing, Communication & Automation*. IEEE, 720–723.
[80] Unal Tatar, Yasir Gokce, and Brian Nussbaum. 2020. Law versus technology: Blockchain, GDPR, and tough tradeoffs. *Comput. Law Secur. Rev.* 38 (2020), 105454.
[81] Sergei Tikhomirov. 2018. Ethereum: State of knowledge and research perspectives. In *10th International Symposium on Foundations and Practice of Security (FPS'17)*. Springer, 206–221.
[82] Viktor Trón, Aron Fischer, Dániel A. Nagy, Zsolt Felföldi, and Nick Johnson. 2016. *Swap, Swear, and Swindle: Incentive System for Swarm.* Technical Report, Ethersphere Orange Papers 1 (2016).
[83] Florian Tschorsch and Björn Scheuermann. 2016. Bitcoin and beyond: A technical survey on decentralized digital currencies. *IEEE Commun. Surv. Tutor.* 18, 3 (2016), 2084–2123.

[84] J. Vijayalakshmi and A. Murugan. 2019. Revamp perception of Bitcoin using cognizant merkle. In *Emerging Research in Computing, Information, Communication and Applications Conference (ERCICA'18)*. Springer, 141–150.
[85] David Vorick and Luke Champine. 2014. *Sia: Simple Decentralized Storage*. Retrieved May 8, 2014. https://sia.tech/sia.pdf
[86] Marko Vukolić. 2016. The quest for scalable blockchain fabric: Proof-of-work vs. BFT replication. In *International Workshop on Open Problems in Network Security (iNetSec'15)*. Springer, 112–125.
[87] Xiaoqing Wang, Chunping Wang, Kun Zhou, and Hongbing Cheng. 2021. ESS: An efficient storage scheme for improving the scalability of Bitcoin network. *IEEE Trans. Netw. Serv. Manag.* 19, 2 (2021), 1191–1202.
[88] Shawn Wilkinson, Tome Boshevski, Josh Brandoff, and Vitalik Buterin. 2014. Storj a peer-to-peer cloud storage network. *White Paper*. https://www.storj.io/storj2014.pdf
[89] Gavin Wood. 2014. Ethereum: A secure decentralised generalised transaction ledger. *Ethereum Project Yellow Paper* 151, 2014 (2014), 1–32.
[90] Junfeng Xie, F. Richard Yu, Tao Huang, Renchao Xie, Jiang Liu, and Yunjie Liu. 2019. A survey on the scalability of blockchain systems. *IEEE Netw.* 33, 5 (2019), 166–173.
[91] Ronghua Xu, Gowri Sankar Ramachandran, Yu Chen, and Bhaskar Krishnamachari. 2019. BlendSM-DDM: Blockchain-enabled secure microservices for decentralized data marketplaces. In *IEEE International Smart Cities Conference (ISC2'19)*. IEEE, 14–17.
[92] Zihuan Xu, Siyuan Han, and Lei Chen. 2018. CUB, a consensus unit-based storage scheme for blockchain system. In *IEEE 34th International Conference on Data Engineering (ICDE'18)*. IEEE, 173–184.
[93] He Xue, Dajiang Chen, Ning Zhang, Hong-Ning Dai, and Keping Yu. 2023. Integration of blockchain and edge computing in internet of things: A survey. *Fut. Gen. Comput. Syst.* 144 (2023), 307–326.
[94] Ruizhe Yang, F. Richard Yu, Pengbo Si, Zhaoxin Yang, and Yanhua Zhang. 2019. Integrated blockchain and edge computing systems: A survey, some research issues and challenges. *IEEE Commun. Surv. Tutor.* 21, 2 (2019), 1508–1532.
[95] Tao Ye, Min Luo, Yi Yang, Kim-Kwang Raymond Choo, and Debiao He. 2023. A survey on redactable blockchain: Challenges and opportunities. *IEEE Trans. Netw. Sci. Eng.* 10, 3 (2023), 1669–1683. https://doi.org/10.1109/TNSE.2022.3233448
[96] Fei Richard Yu and Ying He. 2019. A service-oriented blockchain system with virtualization. *Trans. Blockch. Technol. Applic.* 1, 1 (2019), 1–10.
[97] Fei Richard Yu, Jianmin Liu, Ying He, Pengbo Si, and Yanhua Zhang. 2018. Virtualization for distributed ledger technology (vDLT). *IEEE Access* 6 (2018), 25019–25028.
[98] Changqiang Zhang, Cangshuai Wu, and Xinyi Wang. 2020. Overview of blockchain consensus mechanism. In *2nd International Conference on Big Data Engineering*. 7–12.
[99] Di Zhang, Junqing Le, Xinyu Lei, Tao Xiang, and Xiaofeng Liao. 2021. Exploring the redaction mechanisms of mutable blockchains: A comprehensive survey. *Int. J. Intell. Syst.* 36, 9 (2021), 5051–5084.
[100] Xiaojing Zhang, Rui Qin, Yong Yuan, and Fei-Yue Wang. 2018. An analysis of blockchain-based Bitcoin mining difficulty: Techniques and principles. In *Chinese Automation Congress (CAC'18)*. IEEE, 1184–1189.
[101] Qiuhong Zheng, Yi Li, Ping Chen, and Xinghua Dong. 2018. An innovative IPFS-based storage model for blockchain. In *IEEE/WIC/ACM International Conference on Web Intelligence (WI'18)*. 704–708. DOI: https://doi.org/10.1109/WI.2018.000-8
[102] Qiheng Zhou, Huawei Huang, Zibin Zheng, and Jing Bian. 2020. Solutions to scalability of blockchain: A survey. *IEEE Access* 8 (2020), 16440–16455.
[103] Yan Zhu, Ruiqi Guo, Guohua Gan, and Wei-Tek Tsai. 2016. Interactive incontestable signature for transactions confirmation in Bitcoin blockchain. In *IEEE 40th Annual Computer Software and Applications Conference (COMPSAC'16)*, Vol. 1. IEEE, 443–448.

Received 7 September 2022; revised 17 December 2023; accepted 31 January 2024